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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2024

Code: 101504 | ***Time: 03 Hours***

Semester: V | ***Subject: Hydraulic Engineering*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The boundary layer is defined as the region where
- (i) Pressure forces dominate
- (ii) Inertial forces dominate
- (iii) Viscous forces dominate
- (iv) None of the above

(b) The displacement thickness represents
- (i) Mass loss in boundary layer
- (ii) Momentum loss in boundary layer
- (iii) Velocity loss in boundary layer
- (iv) Reduction in flow rate due to boundary layer

(c) Which of the following is not a parameter of an open channel cross-section?
- (i) Wetted perimeter
- (ii) Pressure head
- (iii) Hydraulic radius
- (iv) Flow depth

(d) In open channel flow, critical flow occurs when the Froude number is
- (i) Less than one (<1)
- (ii) Equal to one (=1)
- (iii) Greater than one (>1)
- (iv) Zero

(e) Chezy's formula is used for
- (i) Pressure flow
- (ii) Uniform open channel flow
- (iii) Uniform flow in pipes
- (iv) Laminar pipe flow

(f) The unit of Manning's roughness coefficient "*n*" is
- (i) m/s
- (ii) Dimensionless
- (iii) $s/m^{1/3}$
- (iv) m^2/s

(g) Which device is used to measure velocity in open channels?
- (i) U-tube manometer
- (ii) Hot-wire anemometer
- (iii) Pitot tube
- (iv) Piezometer

(h) The efficiency of a hydraulic jump is defined by
- (i) Energy gain
- (ii) Energy loss
- (iii) Momentum increase
- (iv) Discharge ratio

(i) Manning's formula is empirical and used to calculate
- (i) Discharge
- (ii) Velocity head
- (iii) Pressure
- (iv) Kinetic energy

(j) A positive surge causes
- (i) Rise in water level
- (ii) Pressure drop
- (iii) Drop in water level
- (iv) Reverse flow

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the development of the boundary layer in open channel flow. Explain the significance of the boundary layer thickness and how it varies along the length of the channel.

(b) A smooth flat plate 1.5 m wide and 2 m long is subjected to flow of water along its length at a uniform velocity of 2 m/s. Assume $\rho = 998$ kg/m^3 and kinematic viscosity $\nu = 10^{-6}$ m^2/s. Find (i) The extent of laminar boundary layer on the plate. (ii) The thickness of boundary layer at edge of laminar boundary layer and trailing edge. (iii) The shear stress at trailing edge.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Define displacement, momentum, and energy thickness. Derive one of them.

(b) A submarine can be assumed to have cylindrical shape with rounded nose. Its length is 60 m and diameter 6 m. Calculate the total power required to overcome boundary friction if it cruises at 10 m/s velocity in sea. Take $\rho = 1030$ kg/m^3 and kinematic viscosity $\nu = 10^{-6}$ m^2/s.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Derive Chezy's equation for uniform flow in an open channel.

(b) A triangular gutter with included angle 60° conveys water at uniform depth of 0.3 m. If the discharge is 0.042 m^3/s. Determine the gradient of trough. Take Chezy's constant = 55.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) What are the geometric parameters of a channel section? Explain with neat diagram.

(b) A trapezoidal channel has side slope 3H:4V and side slope of its bed is 1:2000. Determine optimum dimension of channel. If it carry water at 0.5 m^3/s. Take Chezy's constant = 80.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Define specific energy. Derive the expression for critical depth.

(b) A rectangular channel carries a discharge of 15 m^3/s. The bottom width of the channel is 8 m. The flow depth at a certain section is 1.2 m. Assume the flow is subcritical and frictionless. Calculate: (i) The specific energy of flow at this section. (ii) The critical depth for this discharge. (iii) The critical velocity for this discharge. (iv) The minimum specific energy required to carry this discharge.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain the Direct Step Method for gradually varied flow.

(b) A trapezoidal channel having bottom width = 5 m, side slope = 1:1, bed slope = 1 in 2000, Manning's constant $n = 0.015$, carries 20 m^3/s discharge. If the normal depth is 1.8 m and depth at a section is 2.2 m, compute the length of water surface profile using the direct step method.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Describe different types of hydraulic jumps with illustrations.

(b) In a rectangular channel 3 m wide, water flows at a depth of 0.4 m with a velocity of 6.5 m/s. A hydraulic jump occurs downstream. Calculate: (i) The sequent depth (y_2) (ii) The energy loss in the jump (iii) The length of the jump

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Boundary layer flow in CFD

(b) Specific force and Critical depth

(c) Continuity and energy equation in open channel flow.

(d) Difference between uniform and non-uniform flow.

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